

There is a category of problem in AI development that money alone cannot solve quickly.

You can fund compute. You can hire engineers. You can license existing datasets. But you cannot purchase native deliberative prose in a language where that genre of writing barely exists as a written form. That is the situation with Bangla and chain-of-thought reasoning data.

This article explains why that gap exists, why it matters, and what an open community dataset can do about it.

The Gap That Machine Translation Cannot Fill

Chain-of-thought reasoning — the practice of training models to reason step by step before arriving at an answer — has become a foundational technique in modern AI development. The datasets that support it, however, are overwhelmingly English-origin.

The standard response to this problem for low-resource languages is machine translation: take an English reasoning dataset, translate it into Bangla, and fine-tune on the result. This approach is widely used and produces models that can generate Bangla text. It does not produce models that reason in Bangla.

The distinction matters. Bangla has its own connective tissue of deliberative thought — the phrases and constructions through which a reasoner moves from observation to inference to conclusion. Machine-translated reasoning inherits English scaffolding. The logical connectives come out as calqued English structures. The result reads like dubbed audio: the words are Bangla, but the reasoning voice is not.

A model trained on this kind of data learns to produce Bangla-looking reasoning rather than Bangla reasoning. For low-stakes applications, this may be acceptable. For a public health knowledge system, a clinical decision support tool, or any application where reasoning must be auditable and trustworthy, it is not.

Why the Data Does Not Already Exist

Bangla is spoken by over 230 million people. It is not a marginal language by any demographic measure. Yet its share of deliberative, step-by-step reasoning text on the web is vanishingly small.

This is not because Bangla speakers do not reason. It is because the written genres that predominate — journalism, literature, legal text, religious commentary — do not produce deliberative reasoning as a visible textual artifact. Newspapers assert conclusions. Literature narrates events. Laws declare rules. The step-by-step working-through of a problem, written out explicitly in Bangla, is simply not a well-established written genre.

This means no amount of web scraping will recover what was never written. The data has to be authored from scratch, by people who reason natively in Bangla, writing in a genre they are effectively creating as they go.

What the Dataset Is

The Bangla Chain-of-Thought Dataset is an open, community-built collection of native Bangla reasoning records, each consisting of a self-contained question, an optional context passage, explicit step-by-step reasoning in natural deliberative Bangla, a canonical answer, and annotated wrong paths — plausible incorrect answers with one-sentence explanations of where the reasoning breaks.

Each reasoning step is typed: **recall**, **deduce**, **calculate**, **eliminate**, or **verify**. This process-level annotation supports not only supervised fine-tuning but also preference training, verifier training, and structured evaluation from the same records.

The domains covered reflect both the practical priorities of the Bangladeshi context and the structural requirements of reasoning diversity: public health, law, mathematics, civics, linguistics, general science, daily reasoning, document reasoning, and misinformation rebuttal.

Authorship is per-record and truthful. Items drafted with AI assistance carry a **drafted_by** field identifying the model, alongside the human **author** field identifying the person who corrected and finalized the record. Drafting history is never erased. The entire dataset is released under CC BY 4.0.

The Design Principle Behind Wrong Paths

One aspect of the schema worth explaining is the **wrong_paths** field, which is required rather than optional.

For each item, contributors must document at least one plausible wrong answer and identify precisely which reasoning step breaks and why. This is not a quality assurance mechanism — it is a design choice about what the dataset is for.

A reasoning dataset that only records correct reasoning trains a model to produce correct-looking outputs. A dataset that also records where reasoning goes wrong, and why, provides the material for training models that can recognize and resist specific failure modes. The wrong paths are usable directly as preference pairs and as hard distractors for evaluation.

This reflects a broader principle: the goal is not a dataset of performed intelligence, but a dataset of actual reasoning — including the points at which reasoning fails.

The Connection to Bangla Public Health AI

This project does not exist in isolation from the broader work on Bangla digital health infrastructure described elsewhere on this site.

The Bangla RAG framework for public health — designed to deliver verified, grounded health information in Bangla — requires models that reason reliably in the language. A retrieval system can surface the right passage. It cannot ensure that the model interprets that passage correctly, draws valid inferences from it, or recognizes when a query cannot be answered from the available evidence.

That last layer — reliable reasoning in Bangla — is what this dataset is building toward. The two projects are complementary infrastructure pieces. Neither is sufficient on its own.

An Invitation to Contribute

The dataset is actively growing. It is early stage by design: the intention is long-term accumulation of high-quality records rather than rapid bulk generation of low-quality ones.

Meaningful contribution does not require institutional affiliation or funding. It requires Bangla fluency, subject matter grounding in at least one covered domain, and the willingness to write reasoning carefully — which means one inference per step, plausible wrong paths, and natural deliberative language rather than calqued English scaffolding.

The full schema, authoring rules, and contribution process are documented in the repository.

If you work in public health, law, mathematics education, linguistics, or any of the other covered domains, and you reason in Bangla, your contribution is directly relevant — and currently rare.

GitHub repository: github.com/riazsomc/bangla-cot-dataset

Related Insights

- [Toward a Bangla RAG System for Public Health: Motivation, Architecture, and Early Findings](#)
- [Self-Hosted AI for Health Information: Architecture, Data Sovereignty, and Multi-Audience Access](#)